# OntoTab: A Controlled Study of Ontology-Enriched Text for Tabular Credit Risk

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**Abstract**

An ontology is a curated vocabulary of concepts, definitions, and relations. It could help represent records when different sources describe the same subject in different ways. We test its contribution on 30,000 public Taiwan credit-card records. A frozen BGE-M3 encoder converts each generated subject description into a 1,024-coordinate vector; logistic regression, XGBoost, and a multilayer perceptron learn from these vectors. The primary comparison holds subject facts and text layout fixed while changing neutral context (R2E) to correct ontology context (R5). Average-precision differences R5–R2E are −0.0029, −0.0063, and +0.0008 across the three learners, respectively: this development task shows no positive increment from the ontology-context package. With XGBoost, text-only R5 reaches 0.5069 average precision versus 0.5589 for selected structured fields. A hybrid that adds the structured fields and six rounded ratios to R5 (R9) reaches 0.5568, near the 0.5596 engineered-numeric baseline, without a clear gain over it. The effect varies by learner. These findings motivate keeping numeric inputs in a separate channel. They do not test unseen schemas, genuine customer narratives, or the semantic specificity of ontology relations.

## 1 Introduction

Each row of a credit table records one case to be predicted. We call this a *subject record*; a column gives one attribute, such as credit limit or a monthly bill. The real Taiwan row in Table 1 has a 20,000 TWD credit limit, a 3,913 TWD September bill, and an observed next-month default. The outcome is used to train and evaluate the model, never as an input. Another data source could name or format those attributes differently. An *ontology* is a maintained vocabulary of concepts, definitions, and relations that can make the intended meaning of a mapped field explicit.

We compare *representation arms*: alternative ways to present the same subject records to a prediction model. A structured arm passes column values directly. A text arm first writes a deterministic description, then a pretrained BGE-M3 encoder turns that description into an *embedding*, an ordered list of 1,024 numeric coordinates. The encoder's weights stay frozen; logistic regression, XGBoost, or a multilayer perceptron (MLP) learns from each arm's vectors. Figure 1 shows the routes. A fixed-size embedding accommodates different text content but need not retain every numeric detail.

The central question is whether correct ontology context helps beyond a neutral description of the same facts. R5 and its matched control R2E therefore share observed and permitted derived facts, layout, and a controlled token budget. The comparison changes the context package around those facts; without the missing corrupted-ontology prediction controls it cannot isolate semantic correctness as a mechanism. A separate comparison asks what happens when structured values accompany the text embedding in hybrid arm R9.

This first report uses public Taiwan credit-default data [1], which have structured fields but no independently written customer profiles. We reconstruct the reported measures from aligned prediction arrays and state what this numeric-dominant development setting supports. Row serialization, embeddings, and graph-informed prediction have substantial prior art [5, 6, 14, 20]; our contribution is the controlled context contrast and its bounded result. We relate this result to the broader literature after presenting the experiment.

## 2 Data and experimental design

### 2.1 Task and frozen inputs

We use the public UCI Default of Credit Card Clients dataset, which has 30,000 records and a binary default-payment outcome [1]. The frozen study selects 20 predictors after excluding sex, education, and marital status from model and text inputs. It adds six monthly bill-to-credit-limit ratios for the engineered numeric arm. The outcome is positive for 6,636 subjects (22.12%). The original dataset contains structured numeric and categorical values, with no independent narrative about each customer. Text in our representation arms is generated from these structured facts.

| Public row ID | LIMIT_BAL | AGE | PAY_0 | BILL_AMT1 | PAY_AMT1 | Default next month |
|---|---|---|---|---|---|---|
| 1 | 20,000 | 24 | 2 | 3,913 | 0 | 1 |

Table 1: One actual Taiwan subject record (first public row). Five of its 20 selected predictors are shown. The row ID and observed next-month default are displayed for orientation but are never predictors; sex, education, and marital status are excluded.

Figure 1 follows this row through four principal input routes. The upper path keeps selected columns as numbers. The lower path writes text such as *At 2005-09, [SUBJECT] has credit limit 20000.00 TWD.* The frozen encoder converts the complete description, not one sentence, into a vector. Figure 2 shows the final step shared by all arms.

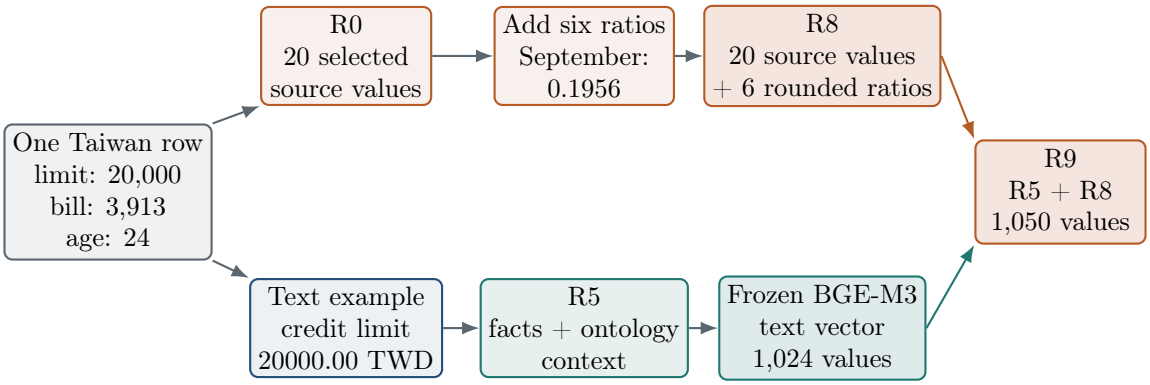



Figure 1: Four routes from one real row to model inputs. R8 appends six rounded ratios; R9 concatenates R8's 26 numeric coordinates with R5's 1,024-coordinate text embedding. The other arms appear in Table 2.

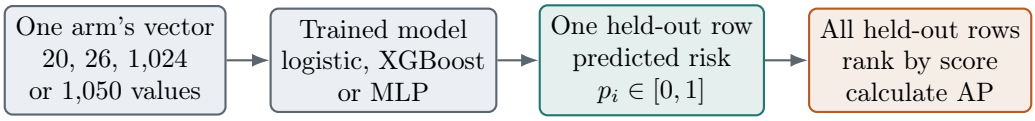



Figure 2: Each arm and learner is fitted separately. Its held-out probabilities are pooled across subjects; average precision (AP) evaluates how well their ranking identifies observed defaulters.

The immutable representation release is `taiwan-r-arms-bge-m3-v1@1.0.0`. A deterministic compiler normalizes and verbalizes observed facts, optional derived facts, and bounded ontology context. The frozen BGE-M3 text encoder yields a 1,024-coordinate vector per subject. No encoder weights are trained on the outcome. The downstream learners are logistic regression, XGBoost, and a small multilayer perceptron (MLP), run through one shared frozen training pipeline.

## 2.2 Representation contrasts

For the row above, September's bill-to-limit ratio is $3\,913/20\,000 = 0.19565$, or 0.1956 after the frozen four-decimal rounding rule. R8 calculates the same kind of ratio for each of six months. More generally, for subject $i$, let $x_i$ contain R0's 20 source predictors: credit limit, age, six repayment-status codes, six bill amounts, and six payment amounts. The codes enter as numeric coordinates without one-hot encoding. R8 is:

$$q_{i,m} = \mathrm{round}_4\left(\frac{b_{i,m}}{\ell_i}\right), \qquad m = 1, \ldots, 6,$$
$$n_i = (x_i, q_{i,1}, \ldots, q_{i,6}) \in \mathbb{R}^{26}. \tag{1}$$

Here $b_{i,m}$ is bill amount `BILL_AMT`$m$, $\ell_i$ is the positive credit limit `LIMIT_BAL`, and $q_{i,m}$ is their rounded quotient. The parentheses in $n_i$ mean that six ratios are appended to the 20 source values. They remain numeric coordinates for the learner; a rounded ratio should not be interpreted as an exact quotient. The frozen numeric matrix is stored as `float32`. Scaling, where used, is fitted within each training fold.

The text arms follow a different route: a deterministic *serializer* writes one description per subject, and the frozen encoder maps that description to 1,024 coordinates. The same input facts and configuration produce the same text; the encoder is not inventing a customer note. The description can contain five distinguishable ingredients: observed facts from selected columns; permitted derived facts computed from those columns; ontology identifiers and readable labels; short concept definitions; and selected hierarchy statements. A property hierarchy can say that one property is a more specific kind of

another. The identifiers, definitions, and hierarchy provide context about *what a field means*; they are not additional observations about the customer. Table 2 shows what happens to the actual Taiwan row in every arm; Table 3 shows short text excerpts emitted by the compiler.

R1 writes source `column=value` pairs. R2 writes the observed values with consistent field names and formatting. R2E and R5 both include those observed facts and the *same* permitted derived facts. In this compiler, the text's credit-utilization derivation uses the September 2005 bill and limit only when unit, account scope, and reference time match. R8's six monthly numeric ratios are a separate engineered baseline; they are not all inserted into R5 text. R2E fills aligned slots with neutral restatements of existing metadata. R3 adds explicit statements linking readable field labels to ontology identifiers; R3D adds definitions; R4 adds bounded `rdfs:subPropertyOf` hierarchy statements without definitions. R5-ND (*no derivations*) combines identities, definitions, and hierarchy; R5 also includes the permitted derived fact. R9 concatenates R5's embedding with R8's 26 numeric coordinates, producing 1,050 coordinates without a second encoder call.

| Arm | What happens to the *same* example subject | Learner input |
|---|---|---|
| R0 | Keep selected typed columns: credit limit 20 000; age 24; September bill 3 913; 17 other values. | 20 source values |
| R8 | Take R0 and append six bill/limit ratios; the September ratio for this row is 0.1956 after rounding. | 26 numeric |
| R1 | Write source pairs such as `LIMIT_BAL=20000`, `AGE=24`, and `BILL_AMT1=3913`. | 1,024 dims |
| R2 | Write the same observed facts with canonical field names and formatting: credit limit 20 000 TWD; September bill 3 913 TWD; age 24 years. | 1,024 dims |
| R2E | Take canonical observed facts, add the permitted September utilization 0.1956 and neutral, length-matched explanatory slots. | 1,024 dims |
| R3 | Take R2 and add explicit statements that connect readable field labels to ontology IDs, such as `otc:creditLimit`. | 1,024 dims |
| R3D | Take R3 and add short definitions of its mapped concepts. | 1,024 dims |
| R4 | Take R3 and add selected property-hierarchy links, *without* the definitions used in R3D. | 1,024 dims |
| R5-ND | Combine R3's identities, R3D's definitions, and R4's hierarchy, but omit the derived utilization fact. | 1,024 dims |
| R5 | Take R5-ND and include the same permitted derived utilization fact as R2E. | 1,024 dims |
| R9 | Place R5's 1,024 embedding coordinates next to R8's 26 numeric coordinates: $[z_i^O; n_i]$. | 1,050 hybrid |
| R6 | Start from R5, then intentionally map selected fields to wrong but compatible concepts; subject values remain. | 1,024 dims |
| R7 | Start from R5, then intentionally rewire selected definition and relation context; subject values remain. | 1,024 dims |

Table 2: One subject through the 11 evaluated arms and two unscored controls (R6/R7). Each text arm reaches the learner as a 1,024-coordinate embedding from the same frozen encoder. R8's numeric input combines source values and six rounded ratios; Table 3 prints text excerpts.

### 2.2.1 A real row, its vectors, and the emitted text

For the first public row, R0's ordered pre-scaling vector and R8's six appended ratios are:

$$\begin{aligned} x_1 &= (20\,000, 24, 2, 2, -1, -1, -2, -2, \\ &\qquad 3\,913, 3\,102, 689, 0, 0, 0, 0, 689, 0, 0, 0, 0) \in \mathbb{R}^{20}, \\ q_1 &= (0.1956, 0.1551, 0.0344, 0, 0, 0), \\ n_1 &= (x_1, q_1) \in \mathbb{R}^{26}. \end{aligned} \tag{2}$$

The first eight entries are credit limit, age, and six monthly repayment-status codes; the next six are bill amounts; the final six are payment amounts. The ratio vector $q_1$ follows the six bill months from September back to April 2005. The entries of $x_1$ are real public-source values; $q_1$ is computed from them, all before fold-local scaling. The public row ID and observed default label are *not* coordinates of $x_1$ or $n_1$.

Table 3 shows how the *same* row looks as text. These are verbatim excerpts from the frozen compiler, not illustrative paraphrases: a full R2 description has 20 observed-fact sentences, and R5 also includes its permitted derived fact and ontology context. The constant token `[SUBJECT]` replaces the source identifier; it is not a customer name. The prefix `otc:` marks an ontology concept identifier, not a value observed for this customer. For example, `PAY_0=2` becomes *repayment status two_ month_ delay.* R6 and R7 examples illustrate their defined text inputs, although their prediction results are not analysed here.

| Arm | Exact excerpt from the first Taiwan row's representation |
|---|---|
| R1 | `LIMIT_BAL=20000; AGE=24; PAY_0=2;` |
| R2 | At 2005-09, [SUBJECT] has credit limit 20000.00 TWD. ⋯ At 2005-09, [SUBJECT] has repayment status two_month_delay. |
| R2E | At 2005-09, [SUBJECT] has derived credit utilization ratio 0.1956. ⋯ The record includes credit limit in its canonical field form. |
| R3 | credit limit is represented by otc:creditLimit. |
| R3D | credit limit: Maximum authorized amount of a revolving credit account. |
| R4 | credit limit specializes credit exposure amount. |
| R5-ND | credit limit is represented by otc:creditLimit. ⋯ credit limit: Maximum authorized amount of a revolving credit account. ⋯ credit limit specializes credit exposure amount. |
| R5 | At 2005-09, [SUBJECT] has derived credit utilization ratio 0.1956. ⋯ credit limit: Maximum authorized amount of a revolving credit account. |
| R6 | original unpaid principal balance is represented by otc:originalUnpaidPrincipalBalance. |
| R7 | credit limit: Principal balance of a loan at origination. |

Table 3: Verbatim excerpts from the first public row. The ⋯ marks omitted sentences and is not emitted text. Each text arm sends its *complete* description to BGE-M3; R0/R8 send numeric vectors, and R9 joins R5's embedding to R8's vector. R6/R7 illustrate unscored wrong-context controls.

The encoder turns the *whole* R5 description into a 1,024-coordinate vector $z_1^O$. Individual coordinates are learned summary features, not direct measurements of the 20,000 TWD limit or 0.1956 utilization. R9 appends $n_1$ to that text vector, $h_1 = [z_1^O; n_1]$, without replacing the source numbers or re-encoding them. We do not print invented embedding coordinates as if they were measured outputs.

Our primary contrast is $\Delta_{\text{context}} = \text{AP(R5)} - \text{AP(R2E)}$: the contribution of the complete ontology-context package relative to matched neutral context. The hybrid diagnostics answer different questions: R9–R5 measures adding the typed numeric channel, while R9–R8 measures adding the text vector to

engineered numeric inputs. R6/R7 wrong-context controls have no submitted prediction results, so semantic specificity is unresolved.

The diagrams show two related operations. In the controlled text comparison (Figure 3), the same subject facts $F_i$ are written in two ways: $T_N$ uses neutral context and $T_O$ uses the correct ontology $\mathcal{O}$. The frozen encoder $E$ turns each complete text into a 1,024-value vector. In the hybrid route (Figure 1), R9 joins the ontology-text vector to the 26 numeric values $n_i$ from Eq. 1:

$$z_i^N = E(T_N(F_i)), \qquad z_i^O = E(T_O(F_i, \mathcal{O})), \qquad h_i = [z_i^O; n_i], \tag{3}$$

Thus $z_i^N$ is R2E, $z_i^O$ is R5, and $h_i$ is R9. The semicolon inside $[z_i^O; n_i]$ means column-wise concatenation: the 1,024 text coordinates are followed by 26 original and engineered numeric coordinates. It does not mean adding vectors or asking the encoder to process the numbers again. The R2E–R5 design controls the fact set, slot layout, and tokenizer-specific length within the frozen tolerance; it does not establish that the resulting prose is identical in style or difficulty.

The vector is an *input*, not the predicted outcome. Each arm and learner combination is trained separately. A text embedding coordinate is a learned summary feature, not a bill amount or a human-readable risk factor; R9 therefore retains a separate channel for typed quantities.

Figure 3 makes the primary comparison more precise. Both rows begin with the same subject facts. The intervention changes the explanatory context around those facts, not the customer's bill, limit, repayment history, or label. Both resulting texts pass through the same frozen encoder and the same kind of downstream learner. The effect we report is a difference in ranking quality after these complete pipelines, not a direct measurement of how much of the ontology the encoder understood.

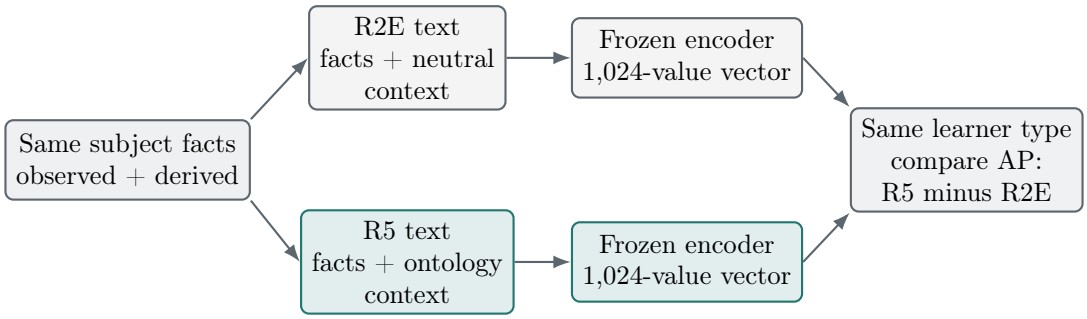



Figure 3: R2E and R5 start with the same facts. Their controlled difference is the neutral versus correct ontology context around those facts; both use the same encoder and learner type. The reported AP difference measures the whole context package, not one ontology relation by itself.

### 2.3 Evaluation and uncertainty

Six submitted experiment bundles were checked for archive integrity, hash binding, unique subject identifiers, finite probabilities in $[0, 1]$, and agreement on labels and five folds of 6,000 records. They cover 11 evaluated arms, three learners, and three seeds (27092031–27092033). For each arm–learner–seed combination, one of the five stratified outer folds is held out. Three stratified inner folds in the remaining 24,000 records choose the highest-mean-AP configuration from four prespecified options; the selected model is refitted on all 24,000 and scores the held-out 6,000.

The options are logistic regularization $C \in \{0.01, 0.1, 1, 10\}$; XGBoost depth $\{3, 6\}$ crossed with learning rate $\{0.03, 0.10\}$ at 300 trees; and MLP hidden layers $\{(128, 32), (256, 64)\}$ crossed with penalty $\alpha \in \{10^{-4}, 10^{-3}\}$. Logistic regression and MLP fit median imputation, missingness indicators, and standard scaling inside each training split; XGBoost uses the supplied numeric matrix directly. Each subject's out-of-fold (OOF) score thus comes from a model fitted without that subject's outer fold. In

Eq. 4, $i$ indexes subjects, $a$ arms, $d$ learners, and $s$ seeds. We average the three aligned OOF *probabilities first*, then rank all 30,000 subjects and compute average precision (AP):

$$\bar{p}_{i,a,d} = \frac{1}{3}\sum_{s=1}^{3} \widehat{p}^{\,\mathrm{OOF}}_{i,a,d,s}, \qquad \mathrm{AP}_{a,d} = \sum_{k=1}^{K} (\mathrm{Rec}_k - \mathrm{Rec}_{k-1})\mathrm{Prec}_k. \tag{4}$$

In the first expression, $\widehat{p}^{\,\mathrm{OOF}}_{i,a,d,s}$ is the predicted default probability from seed $s$, and $\bar{p}_{i,a,d}$ is the three-seed mean for one subject. In the second, we lower a score threshold from highest predicted risk to lowest. At threshold group $k$, precision $\mathrm{Prec}_k$ is the fraction of selected subjects who defaulted, and recall $\mathrm{Rec}_k$ is the fraction of all defaulters selected. The increase $\mathrm{Rec}_k - \mathrm{Rec}_{k-1}$ is the newly recovered fraction of defaulters; AP adds their precision-weighted contributions over the $K$ distinct thresholds, starting at $\mathrm{Rec}_0 = 0$. Thus AP rewards placing actual defaulters near the top of the ranking. This pooled AP is *not* the mean of three seed-level APs. AP is the primary endpoint with a 22.12% positive class; ROC-AUC and Brier score are secondary. We reconstructed the values below from the prediction arrays, without relying on the operators' interpretation files.

A contrast is candidate AP minus baseline AP under the same decoder; a positive value favors the candidate. For example, XGBoost's primary comparison is $0.506909 - 0.513186 = -0.006277$ AP for R5 versus R2E. To describe uncertainty in such paired comparisons, we resample subjects 2,000 times within each observed class. Each resample uses the *same subject indices* for both arms, retaining the pairing, and produces a new AP difference. The reported 95% percentile interval spans the middle 95% of these differences. It describes how this fixed set of OOF scores varies under subject resampling; it does not cover new model fits, a new population, selection among many contrasts, or error from the provisional source bundles. Folds and seeds repeat an evaluation of one dataset; they are not independent studies.

## 3 Results

The main finding is a bounded one: changing matched neutral context to the ontology-context package does not improve AP consistently across learners. The practical hybrid comparison is also learner-dependent. Tables 4 and 5 give the full values and paired differences; Figures 4 and 5 show their scale.

Table 4: Average precision from seed-averaged aligned out-of-fold probabilities on the same 30,000 subjects. Higher is better.

| Representation | Logistic | XGBoost | MLP |
|---|---|---|---|
| R0 raw structured | 0.498312 | 0.558879 | 0.545134 |
| R8 engineered numeric | 0.498109 | 0.559596 | 0.542349 |
| R1 naive text | 0.530716 | 0.520287 | 0.519656 |
| R2 canonical text | 0.531560 | 0.516594 | 0.517589 |
| R2E information matched | 0.525966 | 0.513186 | 0.511484 |
| R3 ontology identity | 0.527561 | 0.512485 | 0.517816 |
| R3D identity and definitions | 0.526062 | 0.516102 | 0.512946 |
| R4 relation context | 0.525773 | 0.515442 | 0.511531 |
| R5-ND no derived facts | 0.521739 | 0.504700 | 0.508618 |
| R5 full ontology context | 0.523069 | 0.506909 | 0.512294 |
| R9 R5 plus typed numeric | 0.528413 | 0.556833 | 0.521824 |

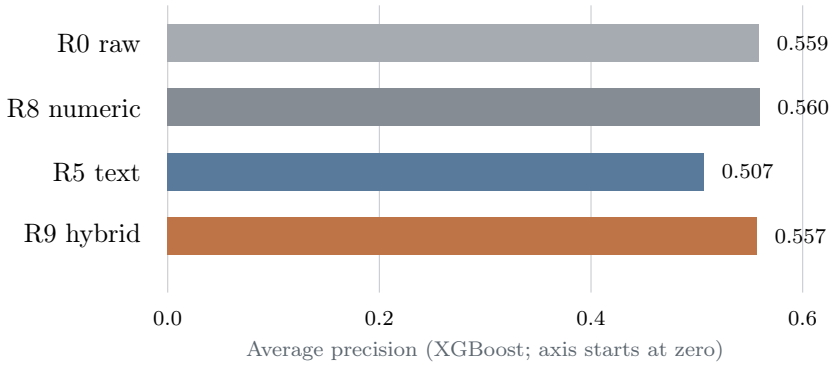



Figure 4: Typed structured values matter for XGBoost. Text-only R5 is below the raw and engineered numeric arms; R9 nearly restores their AP when the R8 values are appended. The small R9–R8 difference is examined with paired intervals in Figure 5.

Table 4 shows no common ordering across decoders. The neutral R2E control is stronger than full R5 for logistic and XGBoost, while MLP differs only slightly in the opposite direction. For XGBoost and MLP, structured R0 outperforms text-only R5 by 0.051970 and 0.032840 AP, respectively (Figure 4). Logistic behaves differently: R5 exceeds R0 by 0.024757 AP, illustrating why a result from one decoder should not be described as a representation-wide property.

Table 5: Paired AP contrasts, candidate minus baseline, with conditional 95% percentile bootstrap intervals. Primary comparison is R5–R2E; hybrid diagnostics are exploratory and unadjusted for multiplicity.

| Contrast | Decoder | Difference | Interval |
|---|---|---|---|
| R5–R2E | Logistic | −0.002897 | [−0.007206, 0.001490] |
| | XGBoost | −0.006277 | [−0.010071, −0.002233] |
| | MLP | +0.000810 | [−0.004675, 0.006400] |
| R9–R5 | Logistic | +0.00534 | [0.00140, 0.00951] |
| | XGBoost | +0.04992 | [0.04320, 0.05658] |
| | MLP | +0.00953 | [0.00421, 0.01527] |
| R9–R8 | Logistic | +0.030303 | [0.023237, 0.037510] |
| | XGBoost | −0.002763 | [−0.006692, 0.001407] |
| | MLP | −0.020525 | [−0.028467, −0.012462] |

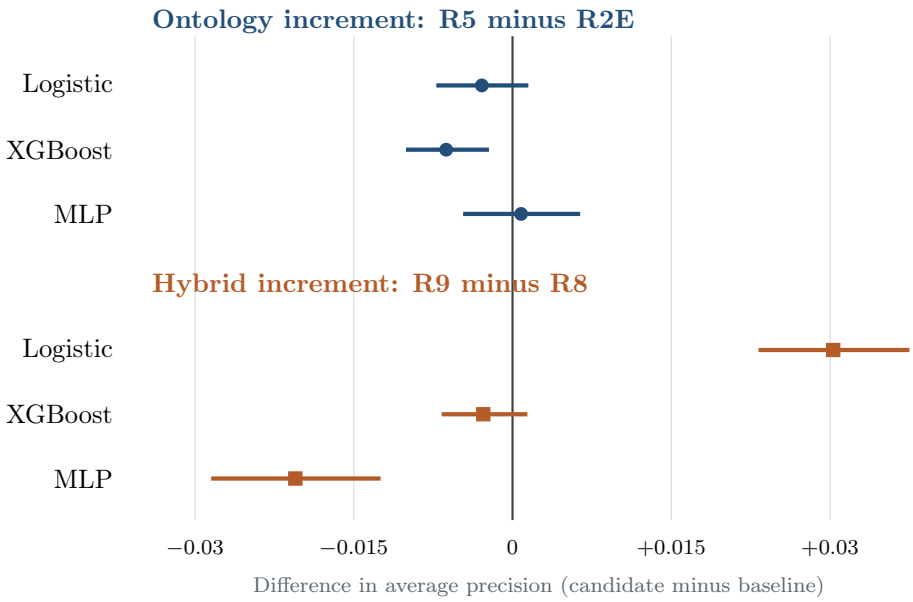



Figure 5: Paired point estimates and conditional 95% subject-bootstrap intervals for the primary ontology contrast (circles) and a hybrid diagnostic (squares). The vertical line is zero. R5 does not show a positive increment over the matched neutral control; adding R5 to numeric R8 has learner-dependent effects. Intervals are unadjusted for multiple comparisons.

The primary differences in Table 5 and Figure 5 show no positive increment from the ontology-context package over matched neutral context in this development setting. XGBoost's paired interval is below zero; logistic and MLP intervals include zero. These estimates neither test equivalence nor identify the benefit of correct relations alone. R6/R7 predictions would be needed to examine semantic specificity.

Appending typed numeric values to R5 has a large effect for XGBoost: R9–R5 is +0.049924 AP. Yet R9–R8 is −0.002763, with an interval crossing zero. Thus the hybrid nearly restores the engineered numeric baseline without demonstrating an added XGBoost benefit from ontology text. Logistic gains from R9 relative to R8, whereas the MLP loses AP. The interaction warrants reporting in full rather than selecting a favorable decoder.

# 4 Positioning against prior work

The evidence above addresses one controlled development comparison. We now place its baselines, text route, and ontology context alongside established approaches, while separating methods tested here from future comparators.

## 4.1 What a competitive tabular baseline must explain

Tabular representation learning now spans specialized models trained on one table, transferable models, and general models applied across tasks [22]. For numeric-dominant prediction, XGBoost and LightGBM remain consequential tree-ensemble references [23, 24]. Empirical comparisons have repeatedly cautioned against assuming that deep tabular models beat tuned trees: architectural rankings depend on datasets, preprocessing, and tuning budgets [25, 2, 3]. TabTransformer shows how contextual embeddings can be learned for categorical features, but this differs from applying a frozen text encoder to a verbalized row [26]. TabPFN, TabICL, and TabDPT instead learn broad tabular priors or scalable in-context prediction, so they are important future comparators for claims about label efficiency or general usefulness [4, 27, 28].

Data quality and evaluation parity also matter: apparent gains can reflect preprocessing or uneven model treatment rather than a semantic mechanism [29, 30].

These studies motivate our structured R0 and engineered R8 controls. R8 adds six ratios to R0, and is therefore a modest feature-engineering baseline, not untouched source data. All three learners use the same frozen local protocol across arms, but this paper does not claim to beat the best tuned XGBoost, LightGBM, or tabular foundation model.

### 4.2 From table content to language representations

Table pretraining work such as TABBIE, TaBERT, and generative table pretraining establishes that column names, cells, and textual context can support transferable representations [31, 32, 33]. TabLLM serializes rows for few-shot classification; TabText uses language-based representations in health prediction [5, 6]. TaRL is particularly close to our frozen-row-embedding pipeline: it pairs language-model representations with a lightweight learner and examines where specialized tabular models retain an advantage [7]. TP-BERTa explicitly represents feature names and relative numerical magnitude, whereas FeatLLM generates features from language-model knowledge [8, 9]. Latte and TabuLa-8B take other routes to importing language priors into tables at different scales [34, 35]; ConTextTab incorporates semantics into tabular in-context learning [15]. Surveys place these choices in a larger family of table-language methods [36].

Mixed structured and authentic text deserves separate treatment. TabSTAR is designed for tables with text fields, and a credit-default study has already compared human-written with AI-refined text [10, 37]. Thus the future customer-profile setting is not a novel modality claim. Our Taiwan data contain no independently authored customer narrative: all R5 text is generated from existing columns. R9 tests whether a typed numeric channel can coexist with that generated representation, not whether genuine notes improve prediction. BGE-M3 supplies the frozen 1,024-coordinate text encoder and was developed for retrieval, not exact financial arithmetic [11].

### 4.3 Cross-table learning and the work of semantic alignment

TransTab handles partially overlapping feature sets across tables; XTab pretrains components for reuse across datasets; CARTE uses row graphs and open-vocabulary features when columns do not match [12, 13, 14]. FLAT studies few-shot generalization across heterogeneous feature spaces, while deep-tabular transfer work asks when learned representations carry between tasks [38, 39]. MediTab consolidates and enriches medical tables; TabGLM jointly uses graph and text views of rows [40, 16]. Together, these methods show that varying columns and graph-structured rows are established research problems, not architectural firsts for OntoTab.

There is also an upstream mapping problem. DTT studies example-driven joinability, stEELlm generates semantic table annotations, and ontology-alignment work validates correspondences between heterogeneous schemas [41, 42, 43]. Biomedical ontology similarity provides an older foundation for measuring relatedness among mapped concepts [44]. SeFNet goes further by linking features across multiple healthcare datasets through semantic feature nets [18]. A governed ontology may help when units, definitions, and source fields vary, but a canonical mapping may suffice; these works make that alternative concrete. The present V0 study uses one Taiwan schema and cannot show unseen-schema transfer, cheaper mapping, or cross-domain generality.

### 4.4 Ontology and knowledge-graph mechanisms

Informed machine learning distinguishes many ways to inject prior knowledge; formal ontologies are only one choice [45]. OWL2Vec* embeds ontology graph, lexical, and logical material, while K-BERT injects knowledge-graph context into language representations [17, 46]. Neuro-symbolic prediction

studies examine possible quality and explanation effects, but ontology coverage can mismatch a specific application [47, 48]. TAROT learns from an LLM-prior feature graph and then refines it for the task, directly challenging any claim that a governed ontology is the only useful semantic graph [19]. GraphCare builds patient-specific knowledge graphs; other clinical work fuses visit sequences with a medical ontology [20, 49]. Their mechanisms and tasks differ from a frozen whole-subject text vector, but they show that ontology and graph context already participate in prediction.

Credit-risk work is also closer than a generic ontology citation suggests. CR-ONTO uses ontology reasoning for credit-default classification, and FinCaKG-Onto models financial knowledge with a causal knowledge graph and domain ontology [21, 50]. Knowledge-graph credit predictors have represented borrower relations, enterprise networks, or loan entities, sometimes alongside XGBoost or other structured learners [51, 52, 53, 54]. A review of machine-learning credit risk documents the wider modeling landscape [55]. These are not head-to-head baselines in this paper, and some model entities or relationships absent from Taiwan; their presence rules out broad novelty claims about graphs, ontologies, or hybrid models for credit.

### 4.5 The boundary of this comparison

These precedents make the contribution narrower than a new row-to-text or graph architecture. R5–R2E measures a complete context package under one frozen encoder and one dataset. R9 separately tests whether keeping typed values changes practical performance. Cross-table transfer, semantic-corruption predictions, genuine narrative input, and mapping cost require new experiments.

## 5 Discussion and limits

The R9 result suggests a specific opportunity for mixed tabular and textual records. With XGBoost, appending the R5 embedding to R8's 26 numeric coordinates yields 0.5568 AP, close to R8's 0.5596, whereas text-only R5 yields 0.5069. The separate numeric channel largely retains the structured baseline for this learner while leaving room for text. The text has no demonstrated added value here: R9–R8 is slightly negative with an interval crossing zero. The MLP also loses AP relative to R8, so near-baseline hybrid performance is not learner-independent.

The more promising use case is a subject record containing both typed fields and genuinely informative text, such as a customer profile or a clinical note written before the prediction time. Such text could carry information absent from the columns, while an R9-like architecture preserves precise structured values. The present Taiwan text is synthesized from those same columns, so this possibility remains a hypothesis rather than an observed benefit. A follow-up should compare a strong structured-only model, structured values plus authentic text with neutral mapping, and structured values plus the same text with matched ontology context. These comparisons would separately test narrative value and ontology-specific value. Datasets with meaningful relations among columns, heterogeneous schemas, or multiple entities could further test the ontology's intended role; the bounded relation context here does not establish it.

A separate operational hypothesis is that a reusable semantic mapping and frozen encoder could reduce the manual feature engineering needed to incorporate a new source with different columns or subject narratives. This would be valuable when the marginal effort of mapping a source to governed concepts is lower than rebuilding a conventional feature pipeline. The present experiment uses one fixed schema and does not measure engineering hours, onboarding time, or end-to-end cost. Ontology curation, mapping maintenance, and embedding inference also impose work. A deployment-oriented comparison should measure time to onboard each new source, human effort, inference cost and latency, and predictive quality against a conventional structured-plus-text pipeline. No reduction to a fraction of the usual preprocessing time is established by these Taiwan results.

The present evidence has six important limits. First, the analysis validates submitted predictions but does not independently retrain every arm from public source. Second, R6/R7 semantic-corruption controls are absent, so semantic specificity is not established. Third, the registered schema-shift views and second domain are absent. Fourth, conditional subject bootstrap intervals exclude uncertainty from refitting and model selection and are not multiplicity-adjusted. Fifth, one historical credit dataset cannot establish transport to other populations or outcomes. Sixth, competitive additional tabular learners and text/graph alternatives remain future baselines. These limits narrow the paper's claim; they do not erase the observed controlled V0 contrast.

The study is a methodological benchmark, not a deployed lending system. Removing direct demographic attributes does not remove all potential proxies or establish fairness. Any use for real credit decisions would require a separate data, calibration, equity, and governance assessment.

## 6 Reproducibility and disclosure

The frozen input release identifier is `taiwan-r-arms-bge-m3-v1@1.0.0`; its manifest SHA-256 is `6676c55c0d628964b0e589e059b517a6ccd47afd7f5139139bcba03caef1b2f6`. The split-manifest SHA-256 is `0218bd65cd16f69d8f4211c8244ecc760dcb1f41675ae153f6845acabf3b09ac`. Headline values were recomputed from the immutable submitted prediction arrays, without using their author-written analysis text. The UCI data source is public [1]; the experiment code and prediction bundles are currently available within the research team, limiting external reruns until a public artifact release is approved. Generative AI assisted language editing and figure-source preparation. The named human authors retain responsibility for the scientific claims, calculations, references, and presentation; AI tools are not authors.

## 7 Conclusion

On Taiwan V0, the matched comparison finds no positive AP increment from the ontology-context package. The XGBoost R9 hybrid nearly recovers the structured baseline by retaining numeric inputs, without improving on that baseline. The next tests are whether authentic pre-cutoff subject text adds predictive value under matched neutral and ontology-enriched controls, and whether semantic mapping reduces new-source integration effort at competitive quality. Neither benefit is established by the present study.

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